EDUCATIONAL MEASUREMENT • VALIDITY • PSYCHOMETRICS • AI IN ASSESSMENT

Sergio Araneda, Ph.D.

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Research scientist in educational measurement and psychometrics (Ph.D., UMass Amherst) working on validity theory, test security, and the use of AI and large language models in assessment. Industry experience at Caveon and College Board; peer-reviewed and preprint research, a Routledge book chapter, and 45+ conference sessions at NCME, ITC, AERA, and COTS.

EXPERIENCE

Research Scientist, Caveon July 2023 – Present (part-time since July 2026)
Remote, from New York City, NY

- Research on randomly parallel testing and on using LLMs with Caveon's SmartItem™ technology for item generation and test security.
- Author and co-author of preprints on LLM-based item generation, reliability of randomly parallel tests, and AI-related test security risks; regular presenter at NCME, ITC, COTS, and ATP.

Associate Psychometrician, College Board October 2022 – April 2023
Remote, from New York City, NY

- Member of the PsyTech Team (Psychometrics): research on item parameter drift and on automated essay scoring for the SAT Essay.

Research Assistant, DEMRE, Universidad de Chile August 2017 – December 2018
Santiago, Chile, and remote from Jersey City, NJ

- Evaluated the use of the "ranking de notas" score in the Chilean university admission system (PSU).

Research Interns: Caveon (2022) and ETS AI Labs (2021) Summers
Remote, from Amherst, MA

- Caveon: built SmartItems and used NLP to generate correct/incorrect statements automatically from text.
- ETS AI Labs: designed feedback logic for filler words and pauses in the "Authentic" interview-practice app.

Earlier career in finance 2013 – 2019
CPQi (Murex consultant for RBC), Murex (Back-Office consultant), Vision Advisors (quant analyst)

EDUCATION

Ph.D., Research, Educational Measurement, and Psychometrics 2019 – 2022
University of Massachusetts Amherst
Dissertation: *An Experiential Approach to Test Design and Validation*.

Mathematical Civil Engineer 2005 – 2013
Universidad de Chile

TECHNICAL SKILLS & LANGUAGES

Programming	Python (since 2013; PyTorch), R (since 2018), SQL, MATLAB, Java, UNIX, Stata, Git
Psychometric software	flexMIRT, Mplus; optimization: Gurobi, CPLEX
Languages	Spanish (native); English (advanced; TOEFL iBT 102/120); French (advanced reading)

SELECTED PUBLICATIONS

Sireci, S. G., Araneda, S., & Suárez-Álvarez, J. (2026). The impact of artificial intelligence on validity and test validation. In *The Role of AI in Assessment* (pp. 312–336). Routledge. doi:10.4324/9781003621522-13

Araneda, S., Fremer, J., & Foster, D. (2024). Commentary: Past, present, and future of educational measurement. *Educational Measurement: Issues and Practice*. doi:10.1111/emip.12632

Araneda, S., Lee, D., Lewis, J., Sireci, S. G., Moon, J. A., Lehman, B., Arslan, B., & Keehner, M. (2022). Exploring relationships among test takers' behaviors and performance using response process data. *Education Sciences*. doi:10.3390/educsci12020104

Araneda, S. (2026). Tests as socio-technical artifacts: A taxonomy for invalidity hypotheses [Preprint]. doi:10.13140/RG.2.2.33888.65285

Araneda, S. (2026). The multiple crises of validity theory in the age of AI [Preprint]. doi:10.13140/RG.2.2.32698.15048

Araneda, S., & Sireci, S. G. (2026). Making the case for an experiential approach to test validation [Preprint]. doi:10.13140/RG.2.2.28455.44968

Foster, C., & Araneda, S. (2026). Assessing pre-exposure risks of LLM-generated item pools [Preprint]. doi:10.13140/RG.2.2.25885.37602

Araneda, S., Marder, A., Foster, C., & Foster, D. F. (2025). Reliability in randomly parallel tests: A practical guide [Preprint]. doi:10.13140/RG.2.2.22375.28329

Araneda, S., Beiting-Parrish, M., Bottoms, B., La Torre, D., et al. (2023). Statement about the new edition of the Standards for Educational & Psychological Testing [White paper]. NCME GSIC. doi:10.31235/osf.io/dv4pa

Full list, including all preprints and newspaper op-eds in *La Tercera*, *El Mercurio*, and *El Desconcierto*: sndxs.github.io/personalwebsite.

SELECTED PRESENTATIONS & SESSIONS

- **ITC 2026, Auckland:** *The Multiple Crises of Validity Theory in the Age of AI*; chaired two symposia (AI in testing and certification; Hispanic-American perspectives on validity).
- **NCME 2026, Los Angeles:** *Factor analysis in randomly parallel tests: Limitations and emerging solutions*.
- **AIME-Con 2026, Pittsburgh (October 6):** *Tests as Socio-Technical Artifacts: A Taxonomy for Invalidity Hypotheses*; chair of the coordinated session on validating AI in educational testing.
- **AIME-Con 2025, Pittsburgh:** training session on generative AI for item construction; organized session on validating AI in educational testing; talks on validity of AI tools and on tests as socio-technical artifacts.
- **COTS 2025 and 2024:** organized a session on AI-driven threats to test security; panelist on generative AI for state assessment programs; SDaR risk-metric talk.
- **NCME 2023–2025; ATP Innovations in Testing 2024–2025; IMPS 2024; ITC 2024; NERA; AERA:** talks and demonstrations on AI item generation, randomly parallel tests, and test security; organizer of NCME 2025 discussion on 100 years of criticism of standardized testing.

SERVICE & TEACHING

- NCME: Website Committee (2025–26); Diversity Issues in Testing Committee (2024–25); Chair, Graduate Student Issues Committee (2021–23), coordinating the Standards Study Group (28 recorded sessions).
- FEVED (Chilean Forum of Educational Assessment Professionals): founding coordinator (2020–21), vice-coordinator (2025–26); organizer of two interview series and a 2025 seminar.
- Teaching: created and taught *Principles of Game Design Applied in Education* (UMass Amherst, Fall 2021); courses in probability and statistics, linear algebra, and mathematical reasoning (Universidad de Chile).